Maku Mazakpe

AI-Native Software Engineer | Mobile, Backend & AI Systems

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PROFESSIONAL SUMMARY

Software engineer with 5+ years delivering production Android, backend, and full-stack systems across fintech and agriculture. Built that engineering foundation before coding agents became mainstream; now applies Codex and Claude Code to accelerate implementation, testing, refactoring, and documentation while retaining human ownership of architecture, code review, security, and releases. Designs and operates an AI research platform with evidence-based quality gates, LLM cost controls, cloud workers, and agent-facing APIs. Experienced in offline-first mobile systems, Kotlin Multiplatform, Python/Django, TypeScript/Next.js, cloud infrastructure, and CI/CD.

TECHNICAL SKILLS

Languages & frameworks: Kotlin, Java, TypeScript, Python, SQL; Jetpack Compose, Kotlin Multiplatform, Next.js, React, Django/DRF, FastAPI

AI engineering: Codex, Claude Code, multi-LLM routing, LangGraph, few-shot retrieval, evidence-grounded research, structured outputs, evaluation, provenance, MCP, agent-facing APIs

Cloud & data: GCP Cloud Run, Cloud SQL/PostgreSQL, Cloud Tasks, GCS, BigQuery; AWS Lambda, Bedrock, DynamoDB; Docker, Vercel, Firebase

Engineering practice: Clean architecture, offline-first sync, REST APIs, security, test-driven development, Git/PR review, GitHub Actions, observability, release automation

EXPERIENCE

Founder & Lead Engineer — StartupTribunal (independent venture) Nov 2025–Present

Accra, Ghana

- Built and operate an AI research platform using TypeScript/Next.js, Cloud Run workers, Cloud SQL/PostgreSQL, GCS report artifacts, Cloud Tasks, and Vercel.
- Implemented multi-model research across Grok, Gemini, Claude, and AWS Bedrock; a five-judge Tribunal, source-provenance rules, few-shot calibration, and approved/cautious/rejected policy govern publication.
- Added LLM cost controls with provider-reported billing ticks, pre-request budget reservations, bounded Retry-After-aware retries, GCS artifact offload, and cached reads; BigQuery analysis helped deliver a historical 41% reduction in monthly infrastructure spend.
- Designed boundaries for paid and high-cost APIs with Firebase server-token verification, server-side entitlements, signed payment webhooks, per-route rate limits, crawler controls, and baseline browser-security headers.
- Built an agent-economy distribution layer: MCP-compatible APIs, feature-gated x402/USDC access, TollBit crawler monetization routing, AI-referral attribution, and schema-backed catalog metadata for search and AI discovery.
- Built a Docker-only release gate covering secret and workflow scanning, strict TypeScript/ESLint, PostgreSQL integration tests, cross-browser/mobile Playwright, and isolated production builds; the Sep 2026 local run passed 363 unit, 61 integration, and 9 browser E2E tests.
- Develop with Codex and Claude Code for scoped analysis, implementation, tests, refactoring, and documentation, retaining PR review, CI checks, production health gates, and human sign-off.

Senior Mobile Developer — Farmerline Group Oct 2024–Present

Accra, Ghana

- Built and modernized production Android applications for agricultural and field operations using Kotlin, Jetpack Compose, Room, WorkManager, Coroutines/Flow, and modular architecture.
- Designed offline-first data flows that preserve pending local changes, replay work after process death, and handle unreliable connectivity, retries, and tenant-scoped data safely.
- Delivered shared Kotlin Multiplatform features across Android and iOS, including native mapping, location, file, and audio integrations.
- Integrated AI-assisted advisory into mobile workflows with conversation history, attachments, offline-aware states, and authenticated APIs.
- Strengthened lifecycle safety, crash prevention, test coverage, Android platform compatibility, CI quality gates, and release automation across mature and greenfield codebases.

Technical Lead — AgriGuard Ltd Dec 2023–Nov 2024

Accra, Ghana

- Architected a farm advisory and insurance backend with Django/DRF, Celery, PostgreSQL, GeoDjango, Docker, and authenticated REST APIs.
- Built AI-supported advisory workflows grounded in agricultural information and geospatial data; integrated secure payment processing.
- Introduced GitHub Actions delivery automation, reducing deployment time 70% versus the prior manual process.

Android Engineer — Accenture Jun 2022–Apr 2023

Nairobi, Kenya

- Developed Android features for financial-services clients, including authentication, secure API integration, automated tests, code review, and Agile delivery.

Full-Stack Developer — Pelel Enterprises Apr 2021–Mar 2022

Arua, Uganda

- Built an agricultural information system spanning an Android client, PHP Lumen APIs, PostgreSQL data model, and JWT authentication.

Android & Web Developer — Freelance and contract 2020–2021

East Africa / Remote

- Delivered Android and web applications, APIs, administration tools, and integrations for small teams and clients.

SELECTED AI PROJECT

StartupTribunal — AI-native startup-validation platform · Founder & Lead Engineer, Nov 2025–Present
Public, production example of multi-model research, evidence-gated publication, agent-facing APIs, and cost-aware cloud delivery. Live product: startuptribunal.com.

EDUCATION & CREDENTIALS

Bachelor of Information Technology — Uganda Christian University

Android Development Certification — Moringa School

Entrepreneurship & Technology Programme — MEST Africa

McKinsey.org Forward Program — McKinsey & Company

Enterprise Design Thinking Practitioner — IBM